Supervyse OPF DCMI User Guide

Revision 1.3

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Revision History

Revision Number	Date	Author	Reviewer	Summary
1.0	2020-03-31	Roy Kuo	Brad Chang	Initial version
1.1	2021-12-27	Roy Kuo	Brad Chang	RV version
1.2	2025-01-22	Jojo Wu	Brad Chang	Move DCMI Porting Guide here

Introduction

The **DCMI spec** defines a baseline set of manageability requirements and interfaces for Data Center Server Management. This document is a guide to make DCMI module of Supervyse OPF BMC firmware working.

Overview

This document lists supported DCMI commands by Supervyse OPF BMC firmware and guides to configure DCMI module of Supervyse OPF BMC firmware. It also provides some methods and some examples to verify DCMI commands after you completed porting it.

Related Information

The following publications and sources of information are referred to by this document or may be useful to you:

- *Data Center Manageability Interface Specification, v1.5 Revision 1.0, Intel, 2011.*
- *Supervyse OPF Platform Porting Guide, 1.70, Insyde Software, 2024*

Terms

The following terms are used throughout this document:

DCGRP

DCGRP represents the use of the Group Extension Network Function code (2Ch) together with DCh as the defining the body code.

DCMI spec

DCMI spec means the *Data Center Manageability Interface Specification, v1.5 Revision 1.0.*

DCMI Porting Guide

Introduction

This section guides to configure related configuration of Supervyse OPF BMC firmware.

Configure Supervyse OPF Firmware XML Configuration

Please refer to the ***Introduction*** of ***Supervyse OPF Firmware XML Configuration*** in the ***Supervyse OPF Platform Porting Guide*** to find the config.xml and then configure the related module of Supervyse OPF BMC firmware.

Configure DCMI module

Below table describes all tags of DCMI module.

Value name	Description	Number	Possible value	Default value
u32Key	An ID for DCMI configuration	1	Fixed	DCMI_ID
u8PowerMgmtSource	Power Management Source: DCMI_PWR_MGMT_NONE means to disable power management. DCMI_PWR_MGMT_BMC means power data comes from BMC. DCMI_PWR_MGMT_BMC means power data comes from ME. **Note**: DCMI_PWR_MGMT_ME is only supported in Intel Archer City platform.	1	DCMI_PWR_MGMT_NONE DCMI_PWR_MGMT_BMC DCMI_PWR_MGMT_ME	DCMI_PWR_MGMT_BMC

Example:

```
<dcmiconfig u32Key="DCMI_ID"
        u8PowerMgmtSource="DCMI_PWR_MGMT_BMC"
/>
```

Configure PMBus module

Please refer to the ***PMBus*** of ***Supervyse OPF Firmware XML Configuration*** in the ***Supervyse OPF Platform Porting Guide*** to configure PMBus module.

Example:

```
<pmbusconfig u32Key="PMBUS_ID"
        …
        u32Ps1InputPowerSensorId="PS1_InputPowerSensor"
        …
        u32Ps2InputPowerSensorId="PS2_InputPowerSensor"
        …
/>
```

Configure Sensors

Please refer to the ***Sensor Porting Guide*** in the ***Supervyse OPF Platform Porting Guide*** to configure related sensors, i.e., inlet temperature sensors, CPU temperature sensors, baseboard temperature sensors and power supply input sensors.

For inlet temperature sensors, CPU temperature sensors and baseboard temperature sensors, their Entity ID and Entity Instance can be checked by the command **ipmitool sdr elist**:

```
ipmitool sdr elist
…
TEMP_INLET1     | 14h | ok  |  55.1 | 25 degrees C
TEMP_INLET2     | 15h | ok  |  55.2 | 24 degrees C
…
TEMP_XXXXX      | C4h | ok  |   7.1 | 38 degrees C
TEMP_XXXXX      | C8h | ok  |   7.2 | 21 degrees C
…
CPU_0_TEMP      | 20h | ok  |   3.1 | 40 degrees C
CPU_1_TEMP      | 22h | ok  |   3.2 | 34 degrees C
…
```

For power supply input sensors, make sure the sensor number of PIA sensor configuration and D-Bus sensor configuration are configured as the same.

Example of D-Bus sensor configuration:

```
{
  "Name": "PSU0_PWR_IN",
  "SensorNumber": "0x3E",
  "Type": "SensorNumConfig"
}
```

Example of PIA sensor configuration:

```
<sensor
    sensor_id="PS1_InputPowerSensor"
    …
    u16IPMISensorNumber="0x3E"
    …
>
  …
</sensor >
```

The reading of power supply input sensors also can be checked by the command **ipmitool sdr elist**:

```
ipmitool sdr elist
…
PSU0_PWR_IN      | 3Eh | ok  |  0.0 | 39 Watts
…
PSU1_PWR_IN      | 3Fh | ok  |  0.0 | 40 Watts
…
```

DCMI Command Support List

Introduction

This table lists all supported DCMI commands of Supervyse OPF BMC firmware.

Support List

DCMI Commands	NetFn	CMD	DCMI 1.5	Supervyse OPF
DCMI Capabilities & Discovery Configuration Commands				
Get DCMI Capabilities Info	DCGRP	01h	M	Y
Get Management Controller Identifier String	DCGRP	09h	M	Y
Set Management Controller Identifier String	DCGRP	0Ah	M	Y
Set DCMI Configuration Parameters	DCGRP	12h	M	Y
Get DCMI Configuration Parameters	DCGRP	13h	M	Y
Platform & Asset Identification Commands				
Get Asset Tag	DCGRP	06h	M	Y
Set Asset Tag	DCGRP	08h	M	Y
Sensor & SDR Commands				
Get DCMI Sensor Info	DCGRP	07h	M	Y
Power Management				
Get Power Reading	DCGRP	02h	O	Y
Get Power Limit	DCGRP	03h	O	Y
Set Power Limit	DCGRP	04h	O	Y

Activate/Deactivate Power Limit	DCGRP	05h	O	Y
Thermal Management				
Set Thermal Limit	DCGRP	0Bh	O	Y
Get Thermal Limit	DCGRP	0Ch	O	Y
Get Temperature Readings	DCGRP	10h	M	Y

DCMI Commands

Get DCMI Capabilities Info

Example:

```
ipmitool raw 0x2c 0x01 0xdc 0x01
dc 01 05 02 00 01 01
```

Verify by below commands:

```
ipmitool dcmi discover
ipmitool raw 0x2c 0x01 0xdc 0x01
ipmitool raw 0x2c 0x01 0xdc 0x02
ipmitool raw 0x2c 0x01 0xdc 0x03
ipmitool raw 0x2c 0x01 0xdc 0x04
ipmitool raw 0x2c 0x01 0xdc 0x05
```

Get Management Controller Identifier String

Example:

```
ipmitool raw 0x2c 0x09 0xdc 0x00 0x0f
dc 0f 42 4d 43 30 30 30 30 30 30 30 30 30 30 30
```

Verify by below commands:

```
ipmitool dcmi get_mc_id_string
ipmitool raw 0x2c 0x09 0xdc 0x00 0x01
ipmitool raw 0x2c 0x09 0xdc [offset] [length]
```

Set Management Controller Identifier String

Example:

```
ipmitool raw 0x2c 0x0a 0xdc 0x00 0x05 0x42 0x4d 0x43 0x30 0x30
dc 05
```

Verify by below commands:

```
ipmitool dcmi set_mc_id_string [mc_id_string]
ipmitool raw 0x2c 0x0a 0xdc 0x00 0x08 0x61 0x6d 0x64 0x2d 0x6f 0x62 0x6d 0x00
ipmitool raw 0x2c 0x0a 0xdc [offset] [length] [data1] ... [dataN]
```

Set DCMI Configuration Parameters

Example:

```
ipmitool raw 0x2c 0x12 0xdc 0x01 0x00 0x01
dc
```

Verify by below commands:

```
ipmitool dcmi set_conf_param activate_dhcp
ipmitool dcmi set_conf_param dhcp_config 0x82
ipmitool raw 0x2c 0x12 0xdc 0x01 0x00 0x01
ipmitool raw 0x2c 0x12 0xdc 0x02 0x00 0x01
```

Get DCMI Configuration Parameters

Example:

```
ipmitool raw 0x2c 0x13 0xdc 0x03 0x00
dc 01 05 01 04
```

Verify by below commands:

```
ipmitool dcmi get_conf_param
ipmitool raw 0x2c 0x13 0xdc 0x01 0x00
ipmitool raw 0x2c 0x13 0xdc 0x02 0x00
```

Get Asset Tag

Example:

```
ipmitool raw 0x2c 0x06 0xdc 0x00 0x04
dc 13 65 78 6d 70
```

Verify by below commands:

```
ipmitool dcmi asset_tag
ipmitool raw 0x2c 0x06 0xdc 0x00 0x01
ipmitool raw 0x2c 0x06 0xdc [offset] [length]
```

Note: Implementation keeps the Asset Tag in the Product Info Area of IPMI FRU data. If the area is not configured, this command will return fail.

Set Asset Tag

Example:

```
ipmitool raw 0x2c 0x08 0xdc 0x00 0x04 0x54 0x65 0x73 0x74
dc 04 05 02 00 01 01
```

Verify by below commands:

```
ipmitool dcmi set_asset_tag a123456789b123456789c123456789d123456789e123456789f123456789g12
ipmitool raw 0x2c 0x08 0xdc 0x00 0x04 0x54 0x65 0x73 0x74
ipmitool raw 0x2c 0x08 0xdc [offset] [length] [data1] ... [dataN]
```

Note: Implementation keeps the Asset Tag in the Product Info Area of IPMI FRU data. If the area is not configured, this command will return fail.

Get DCMI Sensor Info

Example:

```
ipmitool raw 0x2c 0x07 0xdc 0x01 0x40 0x00 0x00
dc 01 01 61 00
```

Verify by below commands:

```
ipmitool dcmi sensors
ipmitool raw 0x2c 0x07 0xdc 0x01 0x40 0x00 0x00
ipmitool raw 0x2c 0x07 0xdc 0x01 0x41 0x00 0x00
ipmitool raw 0x2c 0x07 0xdc 0x01 0x42 0x00 0x00
```

Get Power Reading

Example:

```
ipmitool raw 0x2c 0x02 0xdc 0x01 0x00 0x00
dc eb 03 eb 03 eb 03 eb 03 72 d8 90 66 60 ea 00 00 40
```

Verify by below commands:

```
ipmitool dcmi power reading
ipmitool raw 0x2c 0x02 0xdc 0x01 0x00 0x00
ipmitool raw 0x2c 0x02 0xdc 0x02 0x45 0x00
ipmitool raw 0x2c 0x02 0xdc 0x02 0x5e 0x00
ipmitool raw 0x2c 0x02 0xdc 0x02 0x81 0x00
ipmitool raw 0x2c 0x02 0xdc 0x02 0x86 0x00
ipmitool raw 0x2c 0x02 0xdc 0x02 0xc1 0x00
```

Get Power Limit

Example:

```
ipmitool raw 0x2c 0x03 0xdc 0x00 0x00
dc 00 00 11 f4 01 4e 20 00 00 00 00 0c 00
```

Verify by below commands:

```
ipmitool dcmi power get_limit
ipmitool raw 0x2c 0x03 0xdc 0x00 0x00
```

Set Power Limit

Example:

```
ipmitool raw 0x2c 0x04 0xdc 0x00 0x00 0x00 0x01 0xf4 0x01 0x20 0x4e 0x00 0x00 0x00 0x00 0x0c 0x00
dc
```

Verify by below commands:

```
ipmitool dcmi power set_limit action sel_logging limit 888 correction 50000000 sample 50
ipmitool raw 0x2c 0x04 0xdc 0x00 0x00 0x00 0x11 0xf4 0x01 0x00 0x00 0x00 0x00 0x00 0x00 0x00 0x00
```

Activate/Deactivate Power Limit

Example:

```
ipmitool raw 0x2c 0x05 0xdc 0x01 0x00 0x00
dc
```

Verify by below commands:

```
ipmitool dcmi power activate
ipmitool dcmi power deactivate
ipmitool raw 0x2c 0x05 0xdc 0x01 0x00 0x00
ipmitool raw 0x2c 0x05 0xdc 0x00 0x00 0x00
```

Set Thermal Limit

Example:

```
ipmitool raw 0x2c 0x0b 0xdc 0x37 0x0a 0x20 0x3c 0x40 0x00
dc
```

Verify by below commands:

```
ipmitool dcmi thermalpolicy set 0x40 0x0a disabled disabled sel 0x00 64
ipmitool dcmi thermalpolicy set [Entity ID] [Entity Instance] disabled disabled sel 0x00 64
ipmitool raw 0x2c 0x0b 0xdc 0x40 0x0c 0x20 0x00 0x40 0x00
ipmitool raw 0x2c 0x0b 0xdc [Entity ID] [Entity Instance] 0x20 0x00 0x40 0x00
```

Note: In the section *6.7.2 Set Thermal Limit Command* of **DCMI spec**, thermal management only monitors inlet temperature (entity type 37h and 40h).

Get Thermal Limit

Example:

```
ipmitool raw 0x2c 0x0c 0xdc 0x37 0x0a
dc 32 3c 40 00
```

Verify by below commands:

```
ipmitool dcmi thermalpolicy get 0x40 0x0a
ipmitool dcmi thermalpolicy get [Entity ID] [Entity Instance]
ipmitool raw 0x2c 0x0c 0xdc 0x40 0x0a
ipmitool raw 0x2c 0x0c 0xdc [Entity ID] [Entity Instance]
```

Note: In the section *6.7.1 Get Thermal Limit Command* of **DCMI spec**, thermal management only monitors inlet temperature (entity type 37h and 40h).

Get Temperature Readings

Example:

```
ipmitool raw 0x2c 0x10 0xdc 0x01 0x40 0x00 0x00
dc 02 02 2b 0a 27 0b
```

Verify by below commands:

```
ipmitool dcmi get_temp_reading
ipmitool raw 0x2c 0x10 0xdc 0x01 0x40 0x00 0x00
ipmitool raw 0x2c 0x10 0xdc 0x01 0x41 0x00 0x00
ipmitool raw 0x2c 0x10 0xdc 0x01 0x42 0x00 0x00
```